

LM35 Temperature Sensor & ESP32 Integration

A Comprehensive Technical Breakdown of Hardware Mechanics, ADC Signal Processing, and Code Architecture

1. Overview of the LM35 Temperature Sensor

The **LM35** is a highly precise integrated-circuit temperature sensor manufactured by Texas Instruments. Unlike thermistors (NTC/PTC), whose resistance varies non-linearly with temperature, the LM35 produces an analog voltage output that is directly proportional to the Centigrade (Celsius) temperature scale.

Key Specifications

- **Calibrated directly in:** Celsius (°C)
- **Linear Scale Factor:** +10.0 mV/°C
- **Rated Accuracy:** 0.5°C (at +25°C)
- **Operating Range:** –55°C to +150°C
- **Operating Voltage:** 4V to 30V DC
- **Drain Current:** Less than 60 µA

Pinout Configuration (TO-92 Package)

- **Pin 1 (VCC):** Power supply input (4V to 30V).
- **Pin 2 (VOUT):** Analog output voltage proportional to temperature.
- **Pin 3 (GND):** Ground reference connection.

Working Principle

The LM35 leverages the temperature-dependent forward voltage drop across a bandgap semiconductor junction. The internal signal-conditioning circuitry amplifies this bandgap output to provide a uniform gain scale of **10 mV/°C**.

$$V_{OUT} = 10\text{ mV/}^{\circ}\text{C} \times \text{Temperature (}^{\circ}\text{C)}$$

For example: At 25.0°C, $V_{OUT} = 250\text{ mV (0.250 V)}$. At 100.0°C, $V_{OUT} = 1000\text{ mV (1.0 V)}$.

2. System Circuit & Connection Architecture

The system connects an **ESP32 Microcontroller** to an **LM35 Sensor** and an **SSD1306 OLED Display (128x64)** via I2C communication.

Component	Component Pin	ESP32 GPIO Pin	Description
LM35 Sensor	Pin 1 (VCC)	5V / VIN	Power supply line (LM35 requires minimum 4V for proper operation).
	Pin 2 (VOUT)	GPIO 1 (ADC1_CH0)	Analog output voltage fed to ESP32 ADC channel.
	Pin 3 (GND)	GND	Common ground reference.
SSD1306 OLED	VCC	3.3V	Display power supply.
	GND	GND	Common ground.
	SDA	GPIO 5	I2C Data line (software-configured in code).
	SCL	GPIO 4	I2C Clock line (software-configured in code).

Hardware Note: Although the ESP32 logic operates at 3.3V, the LM35 requires at least 4.0V to operate reliably according to its datasheet. Supplying 5V to the LM35 VCC is recommended. Ensure that the sensor output voltage does not exceed the ESP32's maximum allowed analog input voltage (3.3V). Since 3.3V corresponds to 330°C, the output is inherently safe for the ESP32 under normal operating conditions.

3. ADC Signal Processing & Temperature Calculation

The ESP32 converts the analog voltage output of the LM35 into a digital representation using its internal Analog-to-Digital Converter (ADC).

1. Analog-to-Digital Conversion

The code configures a 12-bit ADC resolution ($2^{12} = 4096$ discrete steps, from 0 to 4095). The reference voltage is defined as $V_{REF} = 3.3 \text{ V}$.

$$\text{Voltage (V)} = (\text{ADC_Value} / 4095.0) \times 3.3 \text{ V}$$

2. Voltage to Temperature Conversion

Given that $10 \text{ mV} = 0.010 \text{ V}$ corresponds to 1°C , the relationship is:

$$\text{Temperature } (^\circ\text{C}) = \text{Voltage (V)} / 0.010 = \text{Voltage (V)} \times 100.0$$

3. Direct Mathematical Expression

Combining both equations yields the unified formula executed by the ESP32:

$$\text{Temperature } (^\circ\text{C}) = [(\text{ADC_Value} / 4095.0) \times 3.3] \times 100.0$$

ESP32 ADC Non-Linearity Consideration: The native ESP32 ADC exhibits non-linear behavior near the upper (3.3V) and lower (0V to 0.1V) extremes. Since LM35 produces low voltages (e.g., **250 mV** at 25°C), calibrating using ESP32's built-in ADC attenuation/calibration APIs (``esp_adc_cal``) or applying a polynomial curve fit can further enhance precision in professional deployments.

4. Detailed Firmware Code Walkthrough

Code Implementation

```

#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// OLED Configuration
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_SDA 5
#define OLED_SCL 4

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// Sensor and ADC Constants
#define LM35_PIN 1
#define ADC_VOLTAGE 3.3
#define ADC_MAX 4095.0

void setup() {
  Serial.begin(115200);

  // Initialize I2C with custom SDA/SCL pins
  Wire.begin(OLED_SDA, OLED_SCL);

  // Initialize OLED at I2C address 0x3C
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED not found!");
    while (true); // Halt execution
  }

  // Set ESP32 ADC resolution to 12 bits
  analogReadResolution(12);

  // Display Boot Screen
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
  display.setCursor(25, 20);
  display.println("LM35 SENSOR");
  display.setCursor(25, 35);
  display.println("Initializing...");
  display.display();
  delay(2000);
}

void loop() {
  // 1. Read Raw ADC Value
  int adcValue = analogRead(LM35_PIN);

  // 2. Compute Pin Voltage
  float voltage = (adcValue / ADC_MAX) * ADC_VOLTAGE;

  // 3. Calculate Temperature in Celsius
  float temperatureC = voltage * 100.0;

  // 4. Serial Debug Output
  Serial.print("ADC: "); Serial.print(adcValue);
  Serial.print(" Voltage: "); Serial.print(voltage, 3);
  Serial.print(" V Temperature: "); Serial.print(temperatureC, 1);

```

```

Serial.println(" C");

// 5. Update OLED Display
display.clearDisplay();
display.setTextCursor(SSD1306_WHITE);
display.setTextSize(1);
display.setCursor(25, 5);
display.println("TEMPERATURE");
display.drawLine(0, 18, 128, 18, SSD1306_WHITE);

display.setTextSize(2);
display.setCursor(10, 32);
display.print(temperatureC, 1);
display.print(" C");
display.display();

delay(1000);
}

```

Step-by-Step Execution Analysis

1. Header Inclusions & Hardware Definitions:

- `Wire.h` handles the I2C protocol communication.
- `Adafruit_GFX.h` and `Adafruit_SSD1306.h` manage rendering graphics, text, and graphics primitives on the OLED screen.
- Custom GPIO pins `SDA=5` and `SCL=4` are specified for ESP32 standard peripheral remapping.

2. Setup Function Initialization:

- `Serial.begin(115200)`; initializes serial debug output at 115200 baud rate.
- `Wire.begin(OLED_SDA, OLED_SCL)`; initializes the I2C bus on the designated GPIO pins.
- `display.begin(...)` configures the OLED driver chip via address `0x3C`. If initialization fails, execution halts in an infinite loop.
- `analogReadResolution(12)`; explicitly sets the ADC bit resolution to 12 bits (**0–4095**).
- A boot screen is displayed for 2 seconds using `display.clearDisplay()`, `display.setCursor()`, and `display.display()`.

3. Sensor Acquisition & Conversion (Loop):

- `analogRead(LM35_PIN)`; samples ADC channel connected to GPIO 1.
- Voltage is computed as $(adcValue / 4095.0) \times 3.3$.
- Temperature is derived using the scale factor **voltage** $\times 100.0$.

4. Display & Serial Output:

- Calculated metrics are formatted and printed to the Serial Monitor.
- The OLED screen is refreshed every 1 second (1000 ms), displaying a title bar, a horizontal divider line rendered via `display.drawLine()`, and an enlarged readout of the current temperature.